Mohamad Mustafa Abdulkadir

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SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL
Frameworks & Libraries: React, Next.js, FastAPI, PyTorch, LangGraph
AI & Machine Learning: LLMs (Training, Fine-tuning, RAG), AI Agents, MCP, LiteLLM, OpenAI/Anthropic APIs
Cloud, Databases & Tools: AWS, PostgreSQL, Snowflake, Hatchet, Power BI, Docker, Nginx

EXPERIENCE

Mansoura Al Ain, UAE
Forward Deployed Engineer October 2025 – Present

- Worked directly with management and operational staff to understand workflows, define requirements, and build and deploy the company's internal operations platform as its sole technical contributor.
- Designed the platform using React, TypeScript, AWS, PostgreSQL, and Snowflake, with role-based access, backend APIs, document management, and scheduled cloud workflows.
- Built and maintained ETL pipelines to ingest company data from AWS into Snowflake for querying and downstream analysis.
- Built dashboards, automated operational reports, and added KPI forecasting to support management decisions.
- Developed an AI agent that matches unidentified rental payments to tenants using bank statements and payment history, reducing reconciliation from roughly two days of manual work by two employees to a few minutes.

ADIA Lab Abu Dhabi, UAE
Machine Learning Research Intern June 2025 – September 2025

- Trained a transformer-based encoder to predict in-hospital mortality from clinical time-series data derived from Electronic Health Records, managing compute-intensive experiments and systematically tracking model performance.
- Built an end-to-end pipeline that turns longitudinal Electronic Health Records into model-ready time series and clinical narratives, cutting projected processing time from 193.4 hours to 18 hours.
- Derived patient-level explanations of the encoder's predictions using attention- and gradient-based attribution, then used them with clinical narratives to fine-tune a transformer decoder for grounded report generation.

Etisalat Dubai, UAE
AI Intern January 2025 – April 2025

- Worked with business stakeholders to define requirements and help integrate an AI-powered recommendation engine into the Starz On Business platform.
- Architected a RAG solution using product guides and FAQs to automate tier-one customer support cases.
- Built a dashboard to track the assistant's performance and key operational metrics.

EDUCATION

United Arab Emirates University January 2021 – May 2025
Bachelor of Science in Computer Science, ***GPA: 3.97 / 4.00*** Al Ain, UAE

SELECTED PROJECTS

German Power Day-Ahead Forecasting Pipeline

- Implemented an end-to-end pipeline integrating 10 ENTSO-E and EEX datasets, with automated ingestion, normalization, deterministic and LLM-generated quality checks, and hourly feature engineering.
- Benchmarked forecasting models using walk-forward validation, achieving a best MAE of 13.65 EUR/MWh. Translated hourly forecasts into weekly and monthly baseload, peak, and off-peak fair-value bands for market-signal analysis.

GNI

- Built an AI agent platform where an orchestrator delegates requests to specialist agents and reviews their output before responding.
- Designed the backend using Python, FastAPI, LangGraph, and LiteLLM, with Hatchet for durable agent execution and PostgreSQL row-level security to isolate user data.
- Integrated external tools through MCP servers and Activepieces, with owner approval before external actions, duplicate-request protection, recovery of interrupted runs after restarts, and per-user usage tracking.

AwesomeRAG

- Built a full-stack document Q&A workspace with document uploads, OCR, vector retrieval, intent routing, and answers that cite their sources.
- Ran Docling OCR and EmbeddingGemma locally, with Gemini as an optional cloud fallback and Qdrant for vector storage.

NutriVision

- Mixed Reality diet-tracking application for HoloLens 2 and iOS that detects food in real time and displays its nutritional content.
- Led the AI component: benchmarked five object detectors, fine-tuned YOLOv11s for food detection, and exported it to ONNX for on-device inference.
- Prototyped a single-image food weight-estimation model on a self-collected dataset; placed third in the UAEU Innovation Challenge and presented the project at Expand North Star Dubai 2024.

PUBLICATIONS

- **DoS-based Fake Task Injection for Disrupted Sensing**, AICCSA, 2025.
- **Vault-PMS: A Vault-Based Password Management System**, IWCMC, 2024.

OPEN-SOURCE CONTRIBUTIONS

Deep-ML

- Contributed a LeetCode-style linear algebra problem on 2D translation matrices, with a guide to representing translation in homogeneous coordinates.